MULTIRATE DSP

Multirate digital signal processing

MULTIRATE DSP

Contents

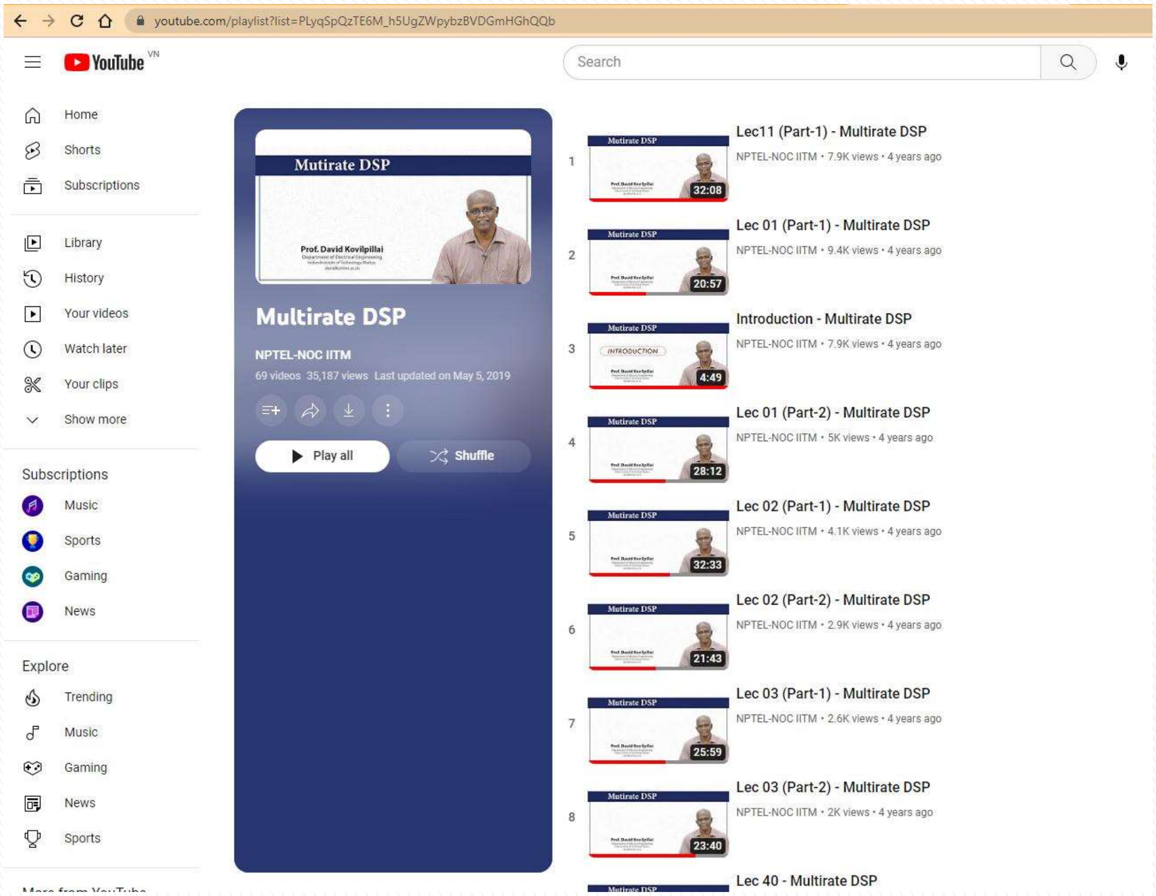
Contents

- https://www.youtube.com/playlist?list=PLyqSpQzTE6M_h5UgZWpybzBVDGmHGhQQb
- https://www.youtube.com/playlist?list=PL4K9r9dYCOopWcmLC0XZqA1Wo-CC8p2ID

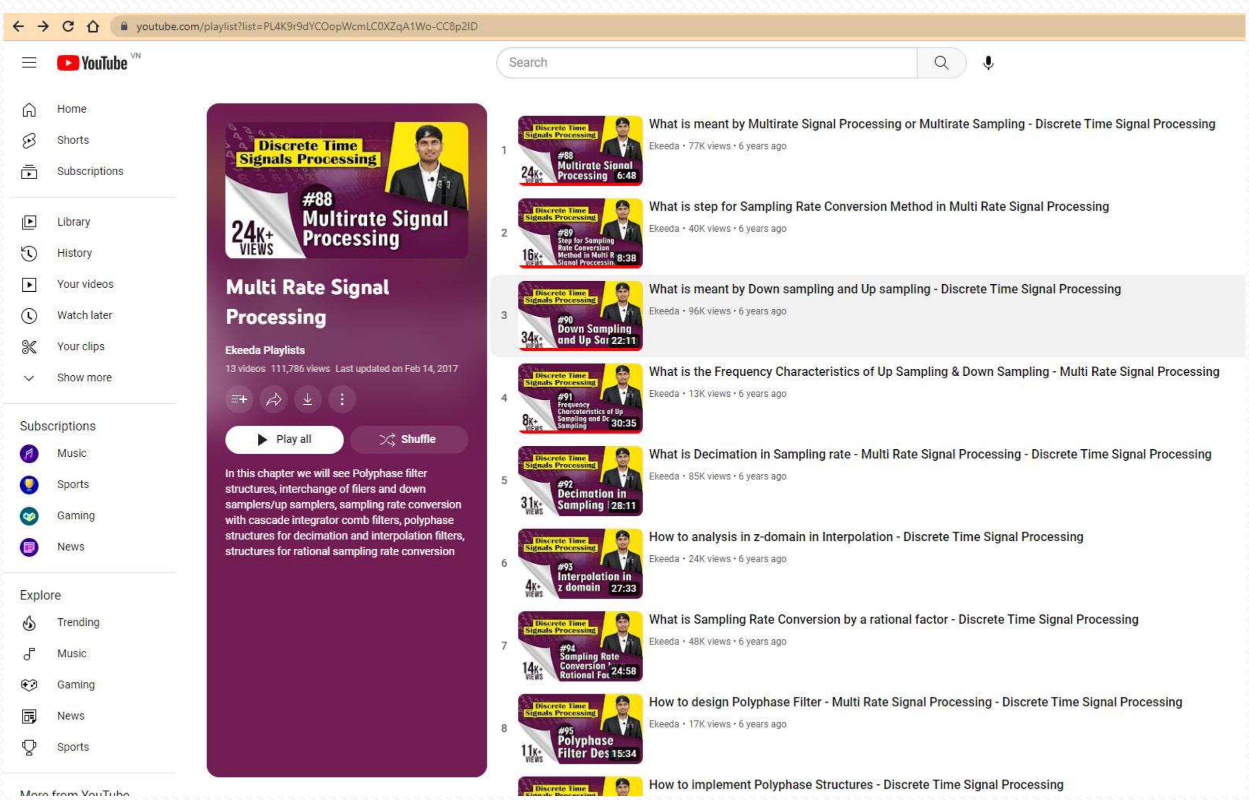
Contents

- Sampling rate conversion (resampling after reconstruction)
  - Upsampling (Interpolation)
  - Downsampling (Decimation)
  - Rational factor
    - several methods for implementing the rate converter: single-stage and multistage
  - Arbitrary factor
- Applications of sampling rate conversion in multirate signal processing systems:
  - narrowband filters
  - digital filter banks
  - subband coding
  - transmultiplexers
  - quadrature mirror filters.

MULTIRATE DSP



MULTIRATE DSP



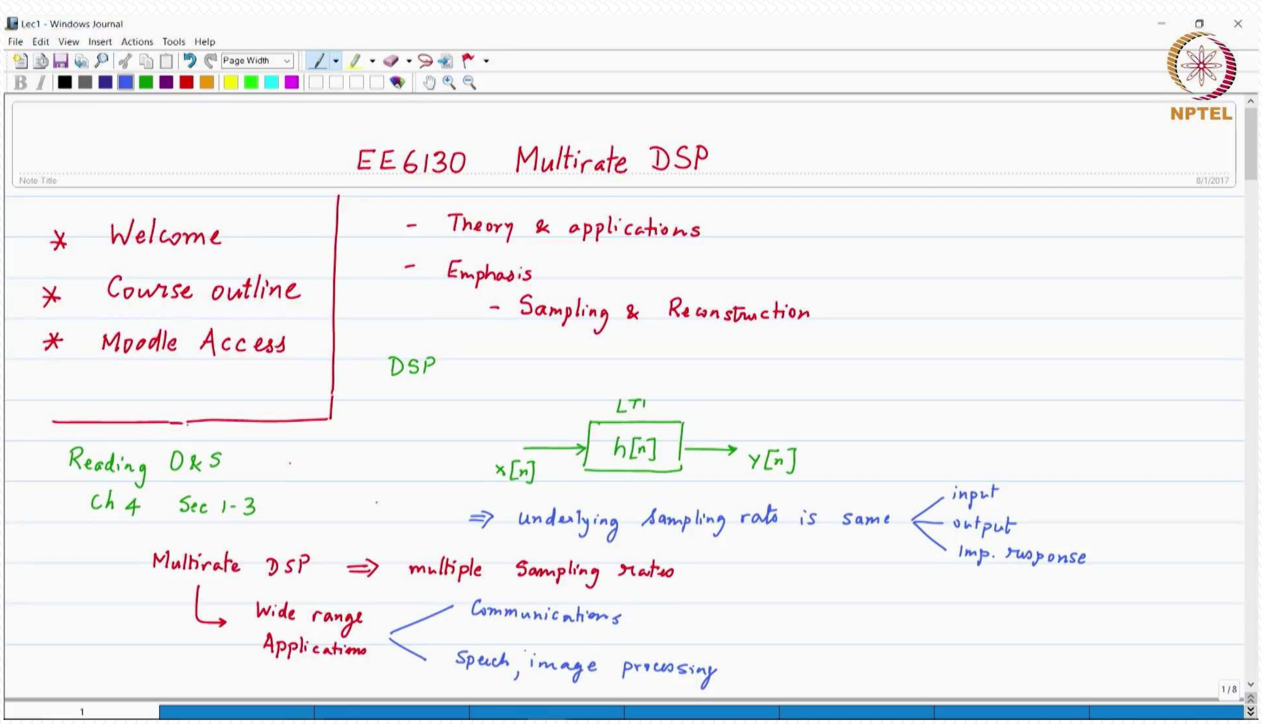
What is multirate DSP?

- In many practical applications of digital signal processing, one is faced with the problem of changing the sampling rate of a signal, either increasing it or decreasing it by some amount.
- For example, in telecommunication systems that transmit and receive different types of signals (e.g., teletype, facsimile, speech, video, etc.), there is a requirement to process the various signals at different rates commensurate with the corresponding bandwidths of the signals.
- The process of converting a signal from a given rate to a different rate is called sampling rate conversion.
- In turn, systems that employ multiple sampling rates in the processing of digital signals are called multirate digital signal processing systems.

MULTIRATE DSP

HHBROS Edu

What is multirate DSP?



MULTIRATE DSP

HHBROS Edu

Example of multirate DSP

Hi-fidelity audio

Human ear 20 Hz - 20,000 Hz

Most sensitive 2-5 KHz

Speech $\leq$ 4 kHz

Music $\sim$ 18 kHz

Broadcast audio 32 KHz

CD 44.1 KHz

DAT 48 KHz

Task

CD $\longrightarrow$ Broadcast audio

44.1 KHz 32 KHz

Two methods for multirate DSP

Hi-fidelity audio

Human ear 20 Hz - 20,000 Hz

Most sensitive 2-5 KHz

Speech $\leq$ 4 kHz

Music $\sim$ 18 kHz

Broadcast audio 32 KHz

CD 44.1 KHz

DAT 48 KHz

Task

CD $\longrightarrow$ Broadcast audio

44.1 KHz 32 KHz

* Complexity
* Analogue components — Filters
* Loss of quality.

Opt 1

CD data $\rightarrow$ D/A (44.1 KHz) $\xrightarrow{\text{analog CT signal}}$ A/D (32 KHz) $\rightarrow$ Broadcast audio

Two methods for multirate DSP

Hi-fidelity audio

Human ear 20 Hz – 20,000 Hz

Most sensitive 2-5 KHz

Speech ≤ 4 kHz

Music ~ 18 KHz

Broadcast audio	32 KHz
CD	44.1 KHz
DAT	48 KHz

Task

CD (44.1 KHz) → Broadcast audio (32 KHz)

Opt 1

CD data → D/A (44.1 KHz) → analog CT signal → A/D (32 KHz) → Broadcast audio

- * Complexity
- * Analogue components — Filters
- * Loss of quality.

Opt 2

CD data → DT proc. → Broadcast audio

$\boxed{\text{Several advantages}}$

Sampling & Reconstruction

* Time $\overset{\mathcal{F}}{\rightleftharpoons}$ Frequency

$x_c(t) \overset{\mathcal{F}}{\rightleftharpoons} X_c(j\Omega)$

CT — CT Fourier FT (CTFT)

$x[n] \overset{\mathcal{F}}{\rightleftharpoons} X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$ (freq.) $\longrightarrow X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi}{N}kn}$, $k = 0, 1, \ldots, N-1$

DT — DTFT: DT, CF — DFT: DT, DF

MULTIRATE DSP

Sampling & Reconstruction

NPTEL

* Time $\xrightleftharpoons{F}$ Frequency

$$x_c(t) \xrightleftharpoons{F} X_c(j\Omega)$$

CT — CT Fourier FT (CTFT)

Sampling ↓ ↑ Reconstruction

$$x[n] \xrightleftharpoons{F} X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

freq.

$$\longrightarrow X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi}{N}kn} \qquad k = 0, 1, \dots N-1$$

DT — DTFT — DFT

DT, CF — DT, DF

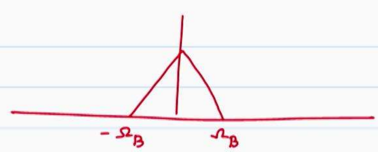
MULTIRATE DSP

Nyquist Sampling Theorem

NPTEL

Bandlimited signal $\quad x_c(t) \xrightleftharpoons{F} X_c(j\Omega)$

$$|X_c(j\Omega)| = 0 \qquad |\Omega| \geq \Omega_B$$



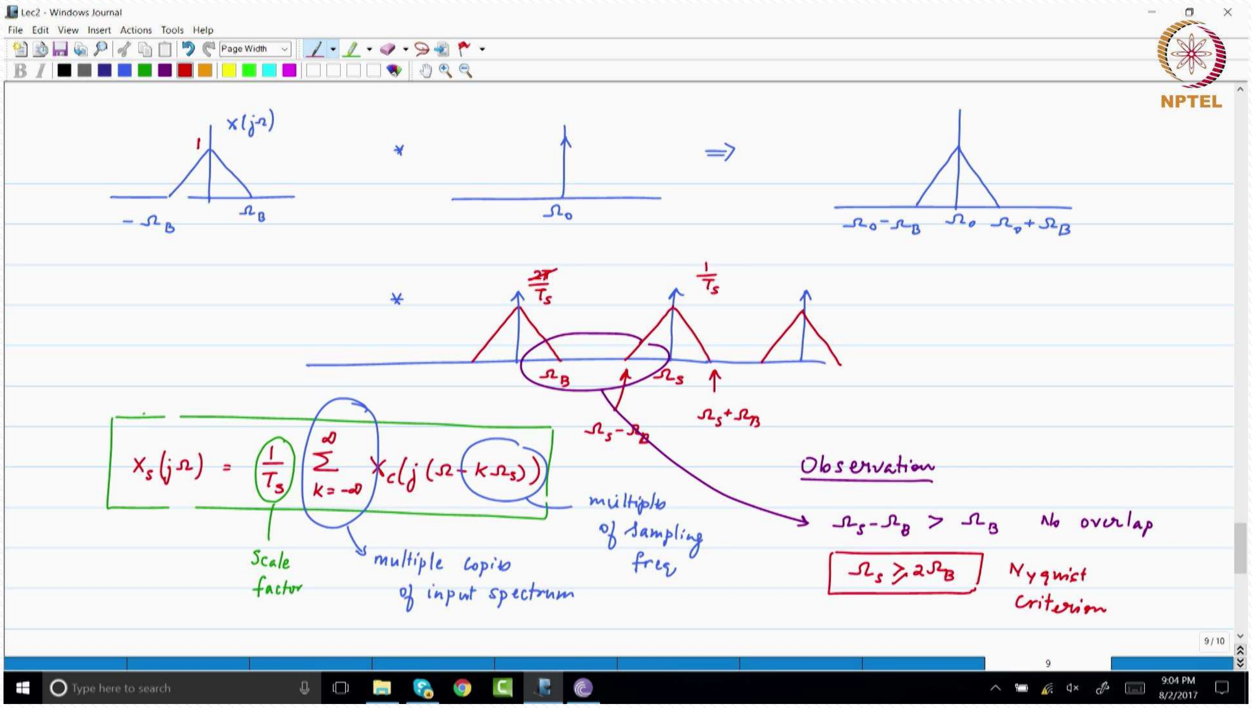
Nyquist Sampling Theorem

If $x_c(t)$ is a BL signal with $|X_c(j\Omega)| = 0 \quad |\Omega| \geq \Omega_B$

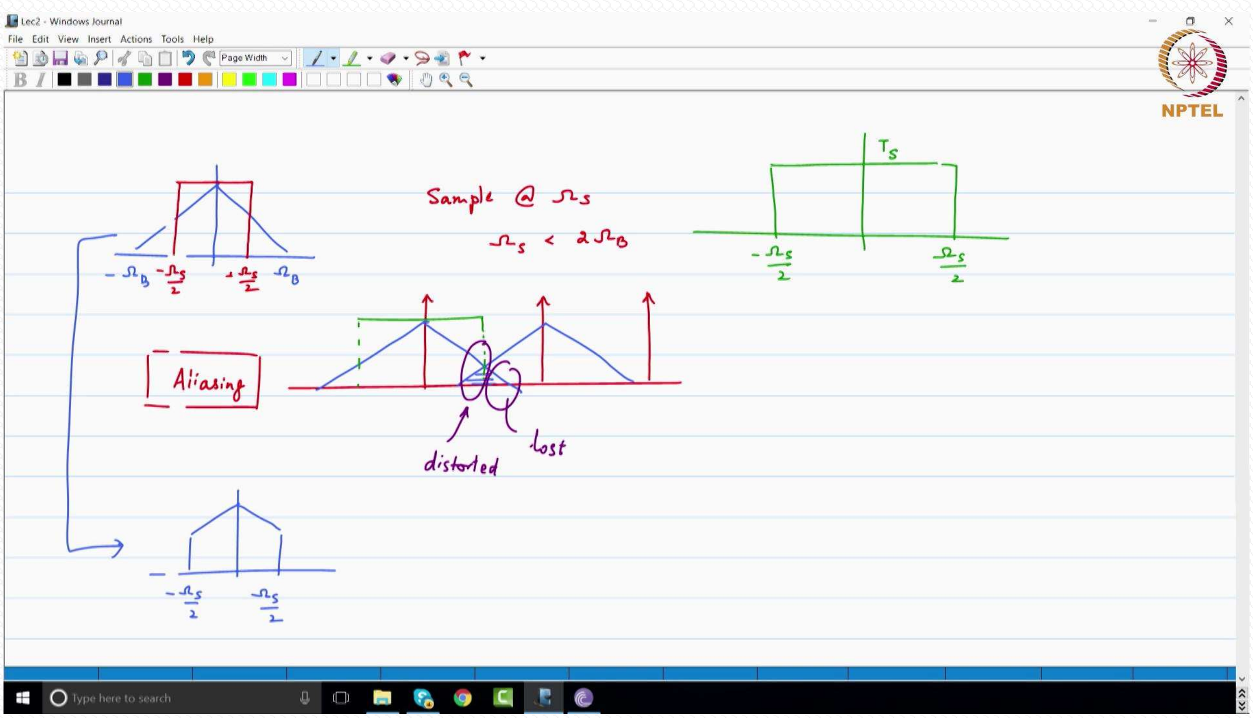
Then $x_c(t)$ is uniquely determined by its samples $x[n] = x_c(nT_s) \quad n = 0, \pm 1, \pm 2 \dots$

if $\Omega_s \geq 2\Omega_B$ & $T_s = \frac{2\pi}{\Omega_s}$

Nyquist Sampling Theorem



Nyquist Sampling Theorem



Upsampling (Interpolation)

Lec7b - Windows Journal

Upsampler / Expander $\quad x[n] \longrightarrow \boxed{\uparrow L} \longrightarrow x_E[n]$

(Increase in Sampling rate by integer factor)

$$x_E[n] = \begin{cases} x\left[\frac{n}{L}\right] & \text{if } n = 0, \pm L, \pm 2L, \ldots \\ 0 & \text{otherwise} \end{cases}$$

$$X_E(z) = \sum_{n=-\infty}^{\infty} x_E[n]\, z^{-n} = \sum_{\substack{n=-\infty \\ = \text{ mult of } L}}^{\infty} x_E[n]\, z^{-n} \qquad n = kL$$

$$\boxed{X(z) = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n}} \quad (1)$$

$$= \sum_{k=-\infty}^{\infty} \underbrace{x_E[kL]}_{x[k]}\, z^{-kL} = \sum_{k=-\infty}^{\infty} x[k]\, z^{-kL} \quad (2)$$

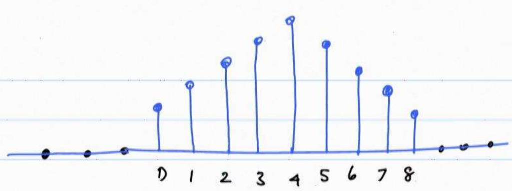
$$\boxed{X_E(z) = X(z^L)}$$

Upsampling (Interpolation)

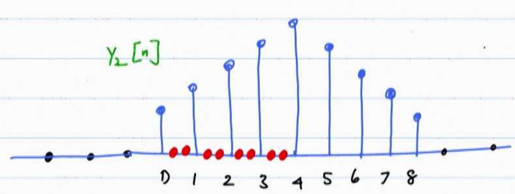
Lec7b - Windows Journal

Expansion

$$Y_2[n] = \begin{cases} x\left[\frac{n}{3}\right] & \text{if } n = 0, \pm 3, \pm 6, \ldots \\ 0 & \text{otherwise} \end{cases} \qquad M = 1,\; L = 3$$



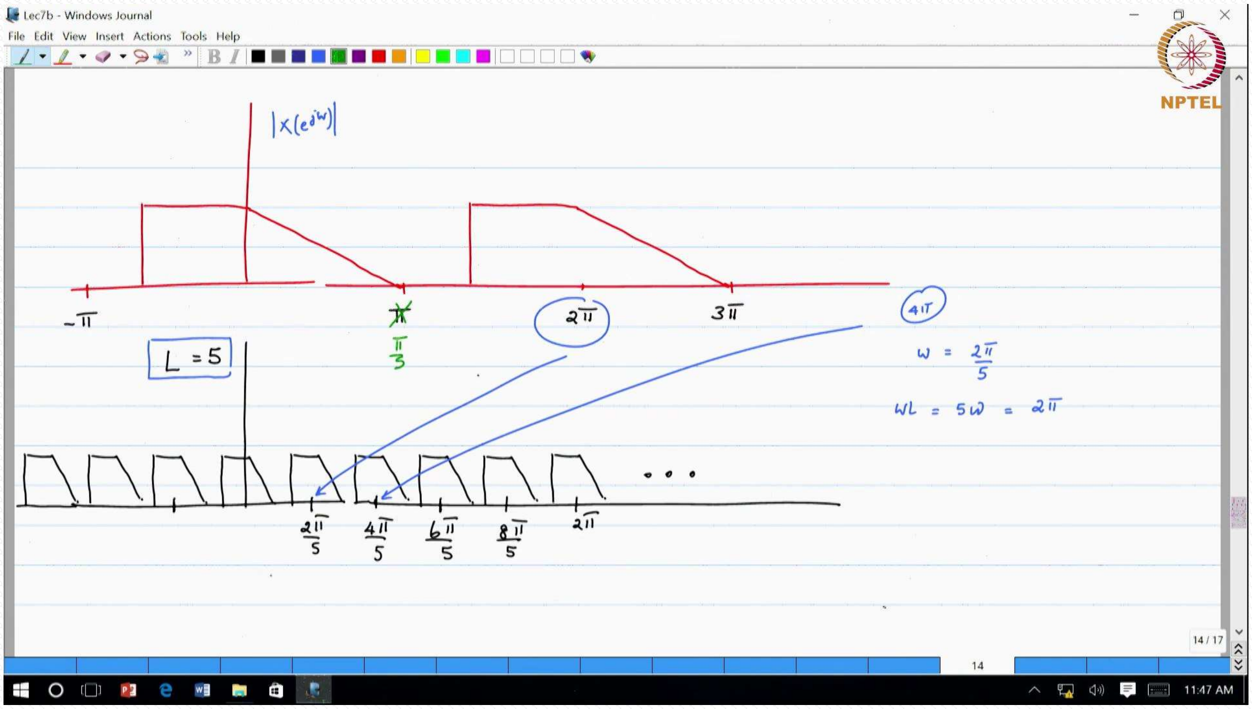
n	0	3
$\frac{n}{3}$	0	1
$x\left[\frac{n}{3}\right]$	$x[0]$	$x[1]$



$$Y_2'[n] = x\left[\frac{n}{L}\right]$$

$\rightarrow$ $(L-1)$ zeros between samples

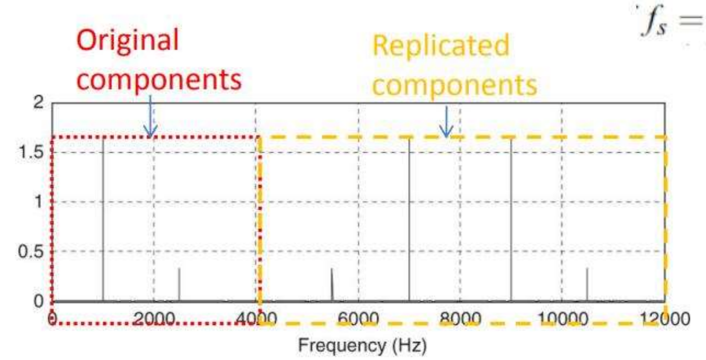
Upsampling (Interpolation)



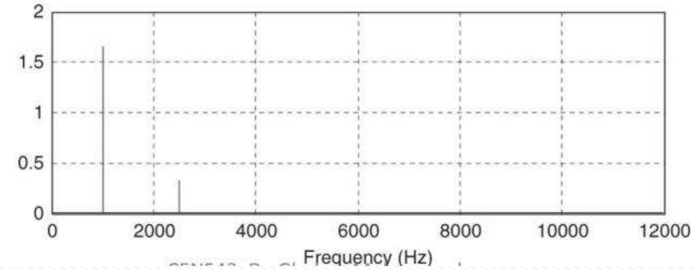
Upsampling (Interpolation)

$$x(n) = 5\sin\left(\frac{2\pi \times 1000n}{8000}\right) + \cos\left(\frac{2\pi \times 2500n}{8000}\right)$$

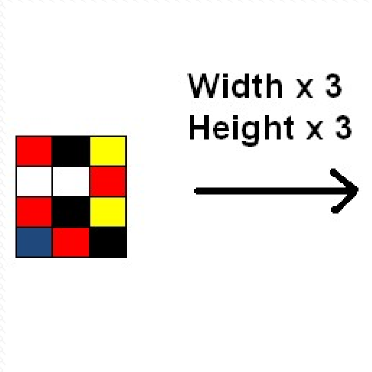
$$f_s = 8{,}000\,\text{Hz}$$

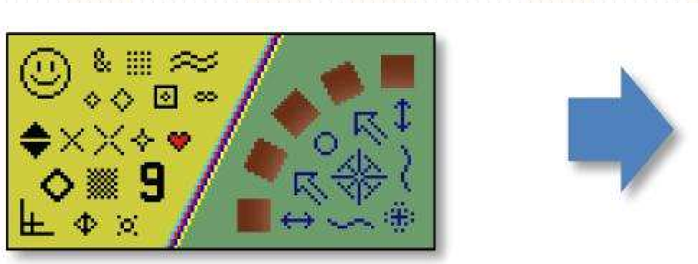


After applying interpolation filter



Upsampling (Interpolation)

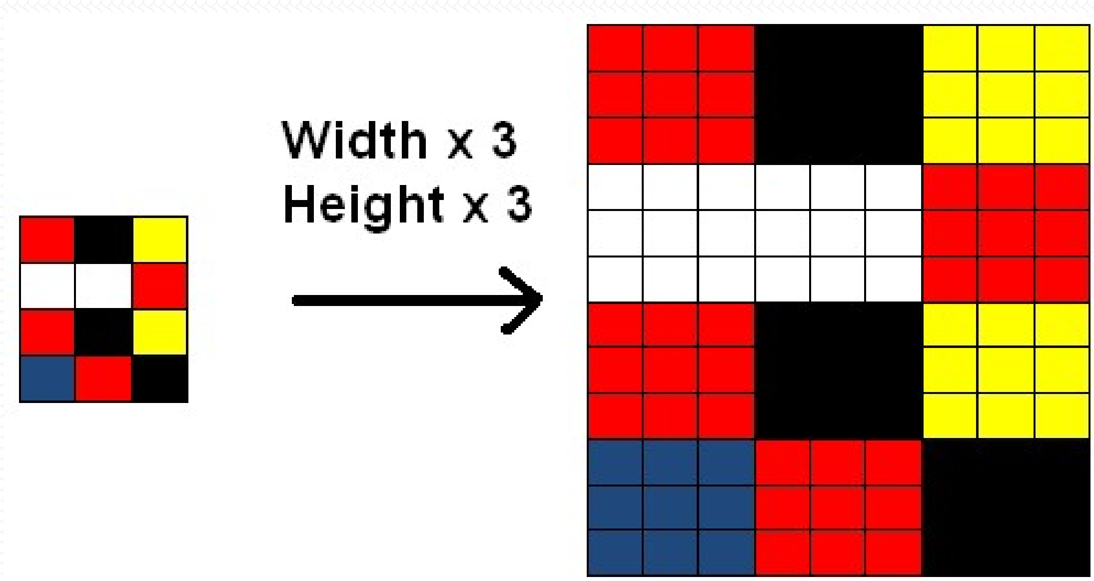




MULTIRATE DSP

HHBROS Edu

Upsampling (Interpolation)



MULTIRATE DSP

HHBROS Edu

Upsampling (Interpolation)

Nearest-neighbor interpolation

hq3x interpolation (ZSNES) http://en.wikipedia.org/wiki/Hqx

Downsampling (Decimation)

Input: x(n). Output: $\boxed{y(m) = x(mM).}$

Taking one sample from every M samples in the input.

Example: $x(n)$: 8 7 4 8 9 6 4 2 −2 −5 −7 −7 −6 −4 . . .

Downsample by a factor of 3

$y(m)$: 8 8 4 −5 −6 . . .

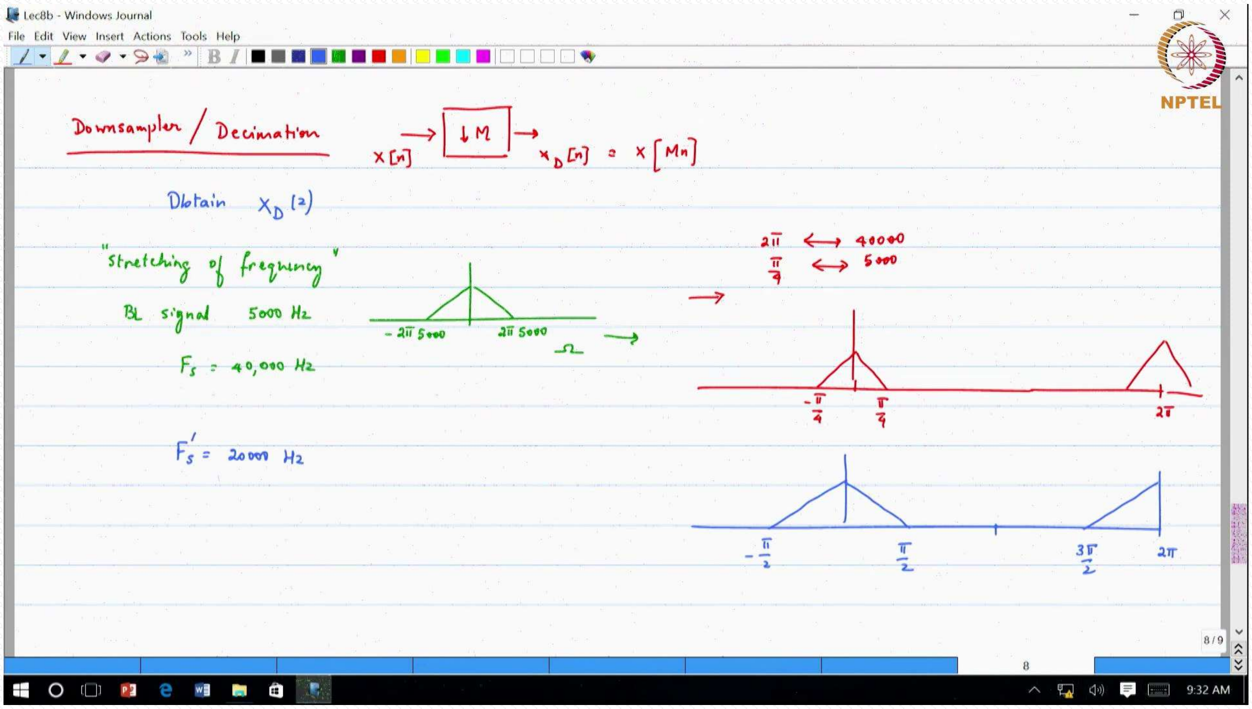
Let, sampling period of x(n) is 0.1 second [10 samples per second]

Sampling period of y(m) is 3×0.1 second [3.3 samples per second]

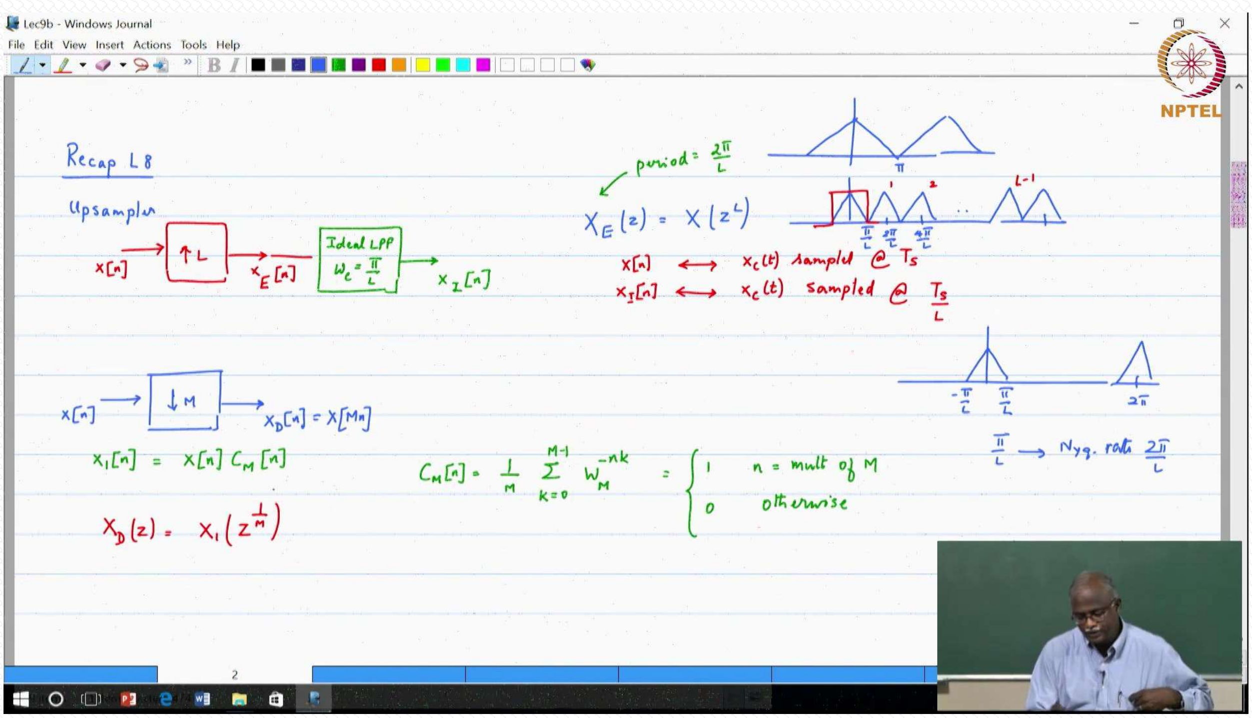
$T \rightarrow MT$

$f_s \rightarrow f_s/M$

Downsampling (Decimation)



Downsampling (Decimation)



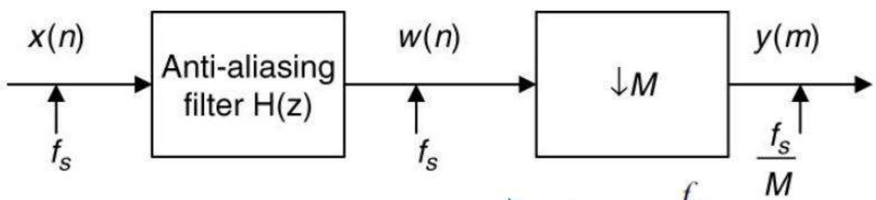
Downsampling (Decimation)

MULTIRATE DSP

Folding frequency after downsampling: $f_{sM}/2 = \frac{f_s}{2M}$

If the signal to be downsampled has frequency components larger than the folding frequency, aliasing noise will be introduced into the downsampled data.

Anti- aliasing filter: H(z) → Low pass filter with stop frequency edge at $f_{sM}/2 = \frac{f_s}{2M}$

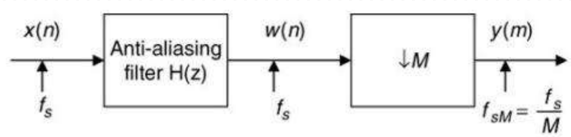


$$\Rightarrow f_{\max} < \frac{f_s}{2M}$$

HHBROS Edu

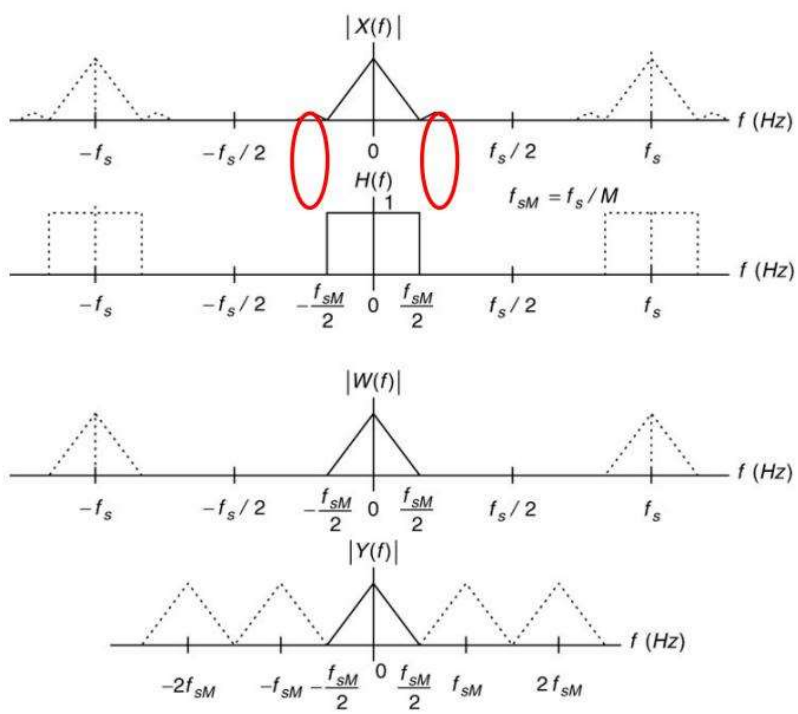
Downsampling (Decimation)

MULTIRATE DSP



Frequency response of anti -aliasing filter ⇒

Output spectrum ⇒

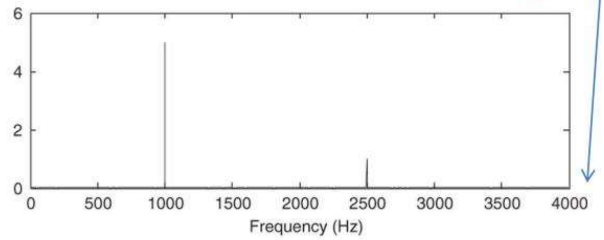


HHBROS Edu

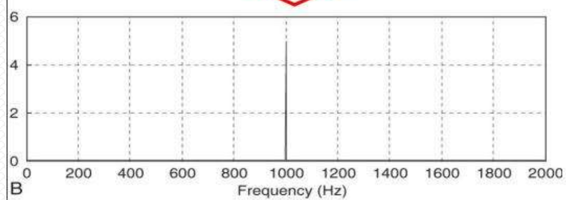
Downsampling (Decimation)

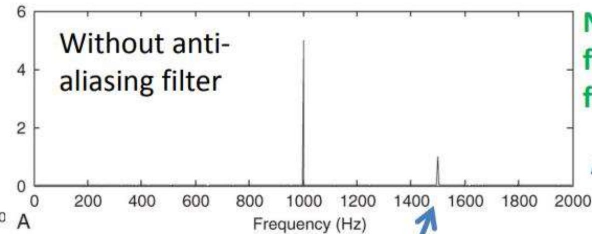
Let: $x(n) = 5\sin\left(\frac{2\pi \times 1000n}{8000}\right) + \cos\left(\frac{2\pi \times 2500n}{8000}\right)$ $f_s = 8{,}000\,\text{Hz}$

Original folding freq.



With anti-aliasing filter





New folding freq.

$4\,\text{kHz} - 2.5\,\text{kHz} = 1.5\,\text{kHz}$ 5

MULTIRATE DSP